

Pundarikaksh N. Tripathi

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Experience

FlyRank AI | Causal ML, X-Learner, Random Forest, scikit-learn, DuckDB, Hugging Face Datasets, Python

Remote

Machine Learning Engineering Intern

July 2026 – September 2026

- Built a refresh-opportunity ranker on 30,000 anonymized pages: dropped label-derived columns (trend_pct, trend_direction), trained a Random Forest on a client-grouped 70/30 split, raising Precision@50 from 0.56 (hand-written rule) to 0.70.
- Quantified leakage: random-split P@50 0.94 vs 0.70 under client-grouped holdout (0.24 optimism gap); reframed claims as observational.
- Capstone (mentor-accepted): X-Learner estimating the CATE of a content refresh (random-forest learners, decile Qini on a client-grouped holdout); ran DuckDB hf:// Parquet queries over the full 79.8M-row warehouse.
- Built an action playbook (reason codes, no-go list); documented the unobserved-confounding limit and the need for an A/B test.

Projects

🔗 **Aegis: Behavioral Governance for Autonomous Agents (OpenAI Buildweek)** | Go, C, eBPF LSM, seccomp, AWS Cedar, SQLite, LinUCB, RL

- Designed and built (one-week sprint; teammates helped with presentation and live testing) a Linux control plane for AI coding agents: C eBPF LSM hooks on file_open, socket_connect and bprm_check_security feed a ring buffer read by a Go daemon.
- Scored event sequences with a temporal graph (zero-trust invariants plus an EMA baseline); flagged events go to a retrieval-augmented adjudicator: exact SHA-256 recall for exec, cosine>0.95 recall otherwise, LLM cascade only on a miss.
- Compiled stored denials (a Cedar forbid subset) into BPF map entries for in-kernel blocking; added a seccomp user-notification exec gate, a Bubble Tea TUI and an offline LinUCB bandit (FN –5, FP –1).
- Validated live on Linux and WSL2 in a recorded demo: blocked the post-exploitation payload of BunnyHijack (a disputed Bun package-manager PoC) at the syscall layer, plus other unauthorised commands.

🔗 [Source: github.com/Quiet-Intelligence/aegis](https://github.com/Quiet-Intelligence/aegis)

🔗 **HiveTorch: Federated Learning over gRPC and Kubernetes** | PyTorch, gRPC, protobuf, safetensors, Kubernetes, Prometheus, FedAvg

- Implemented FedAvg in-process and as a gRPC federation (protobuf, safetensors weights, server-side validation of session, round and tensor shapes); the networked run reproduces the in-process accuracy trace to 1e-6.
- Deployed as Docker Compose services and a Kubernetes Indexed Job (pod index picks the data partition) with Prometheus metrics and alerts; ran 1 server + 5 clients end to end on Minikube.
- Wrote the Dirichlet label-skew partitioner with tests (disjoint cover, determinism, entropy ordering); CI lints, type-checks and runs the container federation.

🔗 [Source: github.com/PundarikakshNTripathi/HiveTorch](https://github.com/PundarikakshNTripathi/HiveTorch)

🔗 **nanoDist: Data Parallelism and ZeRO-2 from Scratch** | Python, NumPy, TCP sockets, ring all-reduce, ZeRO-2, FP16, Hydra, Optuna, W&B

- Implemented ring all-reduce (reduce-scatter + all-gather) in-process and across OS processes over TCP sockets, with bucketed gradients; per-rank traffic verified at $2(N-1)/N$ of the buffer.
- Built ZeRO stage 2 on a from-scratch ring all-reduce: gradient reduce-scatter, Adam on each rank's $1/N$ parameter shard with $1/N$ optimizer state, parameter all-gather; matches single-process Adam (2 and 4 processes).
- 27 tests (Adam equivalence, finite-difference gradients, multi-process TCP); they exposed a crash in the checkpointing path, which I fixed; CI runs ruff, mypy and pytest.

🔗 [Source: github.com/PundarikakshNTripathi/nanoDist](https://github.com/PundarikakshNTripathi/nanoDist)

🔗 **TernixEngine: Ternary GEMM Kernels, CPU (AVX2) and GPU (CUDA)** | C++20, AVX2, CUDA cp.async, PyBind11, GoogleTest, CMake, 1.58-bit

- Wrote branch-free 1.58-bit ternary GEMM in C++20/AVX2: 2-bit packed weights decoded in-register (_mm256_sr1v_epi32, mask, shift) and applied to int8 activations with _mm256_sign_epi32; a 4-row register-blocked variant reuses each decoded weight vector.
- Benchmarked 512^3 on one pinned P-core (i7-14650HX, seeded data): 14.4 ms AVX2 and 8.5 ms register-blocked (31 G ops/s) vs 86.8 ms scalar ($10.2\times$) and 445 ms naive ($52\times$).
- Wrote a 64^3 -tiled CUDA kernel with a two-stage cp.async 16-byte double buffer, bank-conflict-free padded shared rows and 4×4 outputs per thread: 21.3 ms at 4096^3 (6.5 TOPS) on an RTX 5060, $5.6\times$ a naive GPU kernel, bit-exact vs CPU.
- Verified all kernels against an independent integer GEMM (GoogleTest, pytest/NumPy via PyBind11, GPU-vs-CPU exactness); CI builds Linux/Windows and compiles CUDA for sm_80-90. Kernel library in active research and development.

🔗 [Source: github.com/Quiet-Intelligence/TernixEngine](https://github.com/Quiet-Intelligence/TernixEngine)

Achievements

- **3rd on the public and private leaderboards (89,000+ teams), Amazon ML Challenge 2026** at the last check; macro-F0.5 0.991762. Led planning, coordination and review of a 3-person team's entity-resolution pipeline (1.73M businesses vs ≈ 10 M noisy records): 13 lexical retrievers + e5 kNN, LightGBM $A \rightarrow B \rightarrow C$ cascade with cross-encoder blend, calibration, expected-F0.5 set selection.
 - Diagnosed the validation/leaderboard gap (0.9896 vs 0.981) as unseen-country shift (France appears only in test); fixed with per-country IDF and pseudo-labels, adopting changes only if leave-one-country-out improved both ways.

🔗 [Source: github.com/PundarikakshNTripathi/present-day-ber-amlc-2026](https://github.com/PundarikakshNTripathi/present-day-ber-amlc-2026)

- **2nd place, Masai × NoBroker HackArena 2025:** CivicAgent, an AI civic-complaint triage platform (Gemini Vision classification, APScheduler SLA tracking with email escalation, Supabase/PostGIS); proof of concept, no longer hosted.

Technical Skills

Languages: Python, C++20, CUDA C++, PTX (inline asm), Go, C, SQL, TypeScript, Bash

ML Frameworks: PyTorch, JAX, Transformers (Hugging Face), ONNX, Triton, scikit-learn, LightGBM, EconML, DoWhy

Systems and Infra: eBPF (LSM), AVX2 SIMD, CUDA (cp.async, CUB), DDP / ZeRO / ring all-reduce, gRPC, Docker, Kubernetes, CMake, GitHub Actions

Data and Backend: DuckDB, PostgreSQL, SQLite, FastAPI, Supabase, MLflow, Prometheus

Education

Babu Banarasi Das University

Lucknow, UP

B.Tech, Computer Science and Engineering (Artificial Intelligence)

Expected August 2027

- **Coursework:** Data Structures, Algorithm Design and Analysis, Database Management Systems, Computer Architecture, Compiler Design, Theory of Computation.
- **Self-study:** modern C++, CUDA, ML systems, causal inference.